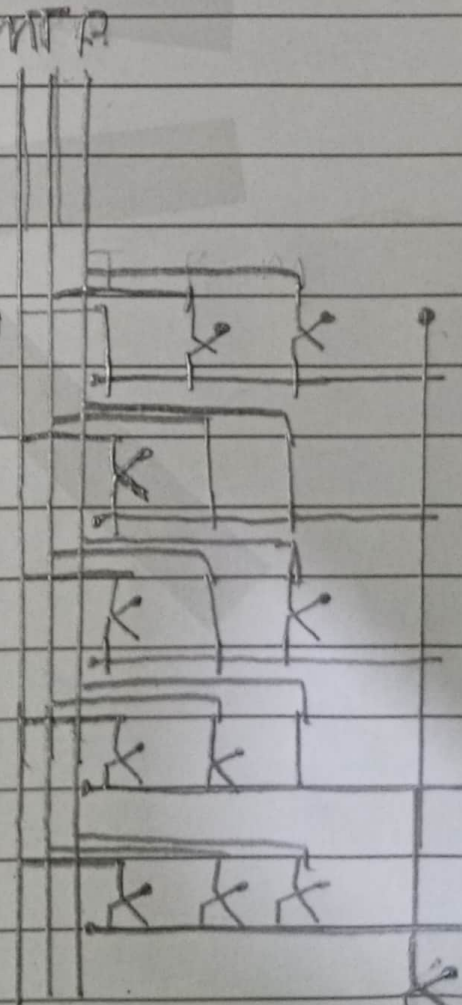
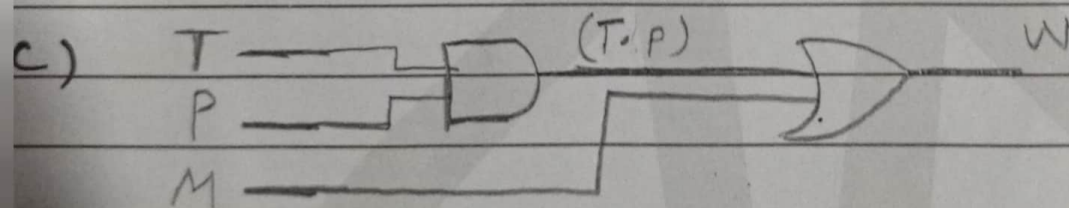


Assignment 3

Q1. a) M = master switch
T = Temperature sensor
P = Pressure sensor
W = Warning light

M	T	P	W	
0	0	0	0	
0	0	1	0	
0	1	0	0	
0	1	1	1	$\bar{M} \cdot T \cdot P$
1	0	0	1	$M \cdot \bar{T} \cdot \bar{P}$
1	0	1	1	$M \cdot \bar{T} \cdot P$
1	1	0	1	$M \cdot T \cdot \bar{P}$
1	1	1	1	$M \cdot T \cdot P$

b) $W = M + (\bar{M} \cdot T \cdot P)$



Q2.a) $W \Rightarrow 1$ if water level > 10 cm, 0 if water level ≤ 10

$R \Rightarrow 1$ if heavy rain is expected, 0 if not

$B \Rightarrow 1$ if fully charged, 0 if $< 20\%$ charged

$P \Rightarrow$ sump pump control, 1 if activated, 0 if not.

W R B P

0 0 0 0

0 0 1 0

0 1 0 0

0 1 1 1 $\bar{W} \cdot R \cdot B$

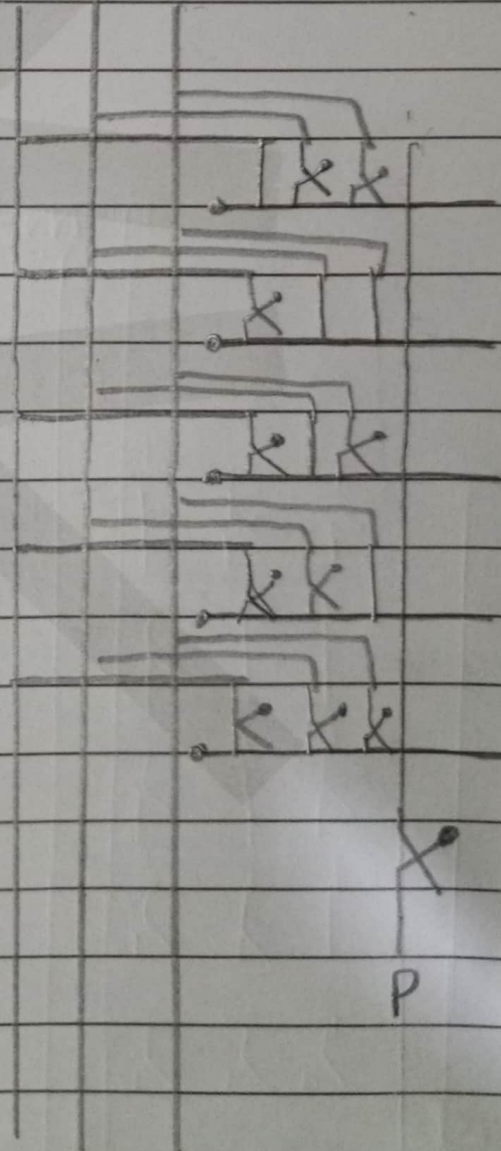
1 0 0 1 $W \cdot \bar{R} \cdot \bar{B}$

1 0 1 1 $W \cdot \bar{R} \cdot B$

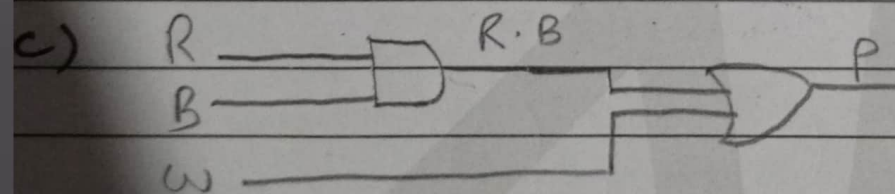
1 1 0 1 $W \cdot R \cdot \bar{B}$

1 1 1 1 $W \cdot R \cdot B$

W R B



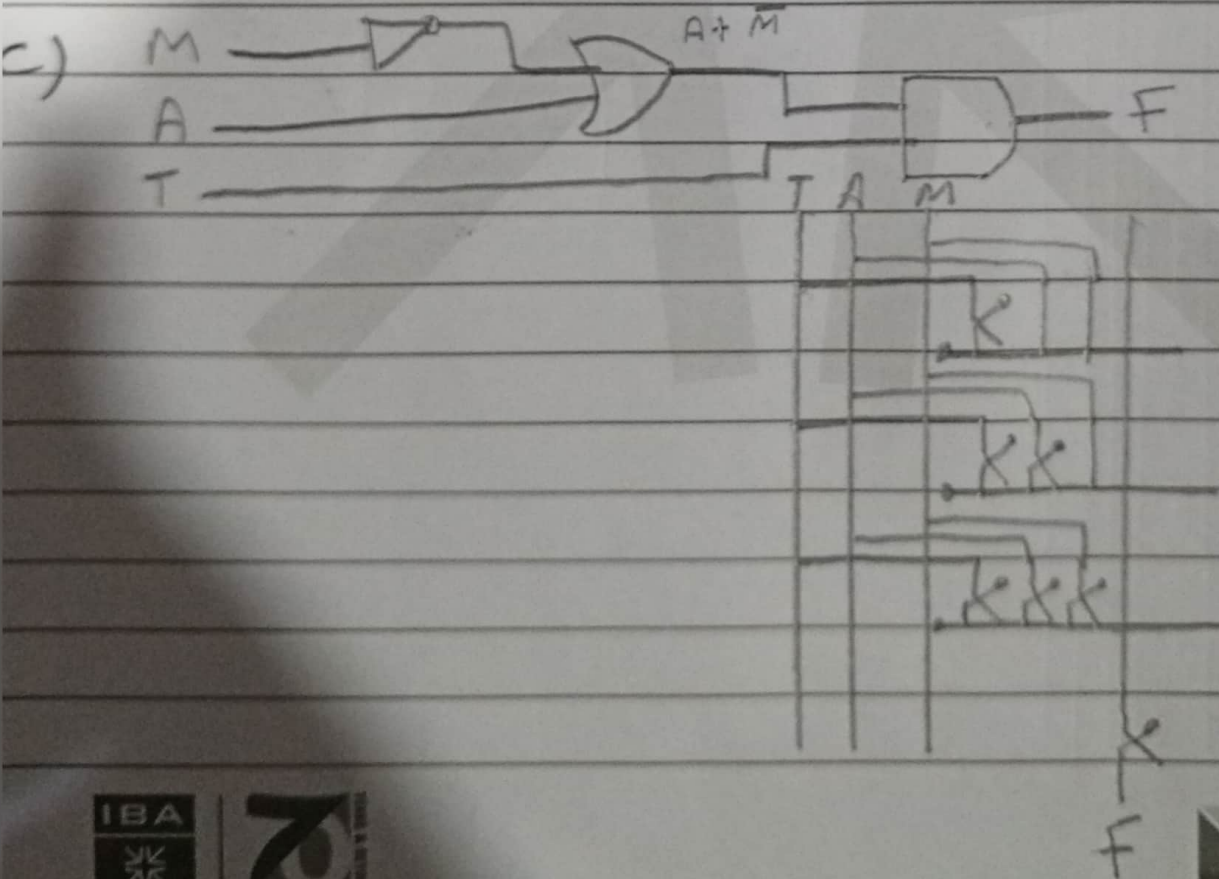
b) $P = W + (\bar{W} \cdot R \cdot B)$



- Q3: $T \Rightarrow 1$ if server temperature $> 80^\circ\text{C}$
 $A \Rightarrow 1$ if room temperature $> 25^\circ\text{C}$
 $M \Rightarrow 1$ if active
 $F \Rightarrow 1$ if ON

T	A	M	F	
0	0	0	0	
0	0	1	0	
0	1	0	0	
0	1	1	0	
1	0	0	1	$T \cdot \bar{A} \cdot \bar{M}$
1	0	1	0	
1	1	0	1	$T \cdot A \cdot \bar{M}$
1	1	1	1	$T \cdot A \cdot M$

b) $F = (T \cdot \bar{M}) + (T \cdot A)$



Q4.a) $T \Rightarrow$ 1 if time between 9 AM and 5 PM

$M \Rightarrow$ 1 if inserted and turned

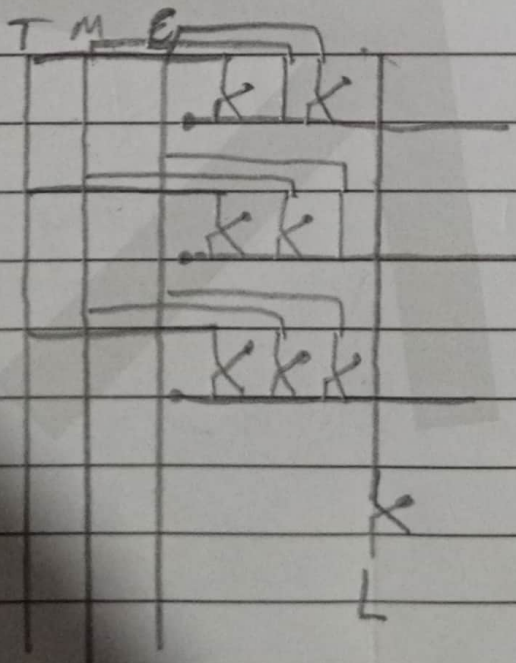
$C \Rightarrow$ 1 if inserted and turned

$L \Rightarrow$ 1 if unlocked

T	M	C	L
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

b)
$$L = (T \cdot \bar{M} \cdot C) + (T \cdot M \cdot \bar{C}) + (T \cdot M \cdot C)$$

c)



Q5. a) $T \Rightarrow 1$ if temperature < 4

$H \Rightarrow 1$ if humidity > 70

$M \Rightarrow 1$ if pressed

$D \Rightarrow 1$ if activated

M T H D

0 0 0 0

0 0 1 0

0 1 0 0

0 1 1 1 $\bar{M} \cdot T \cdot H$

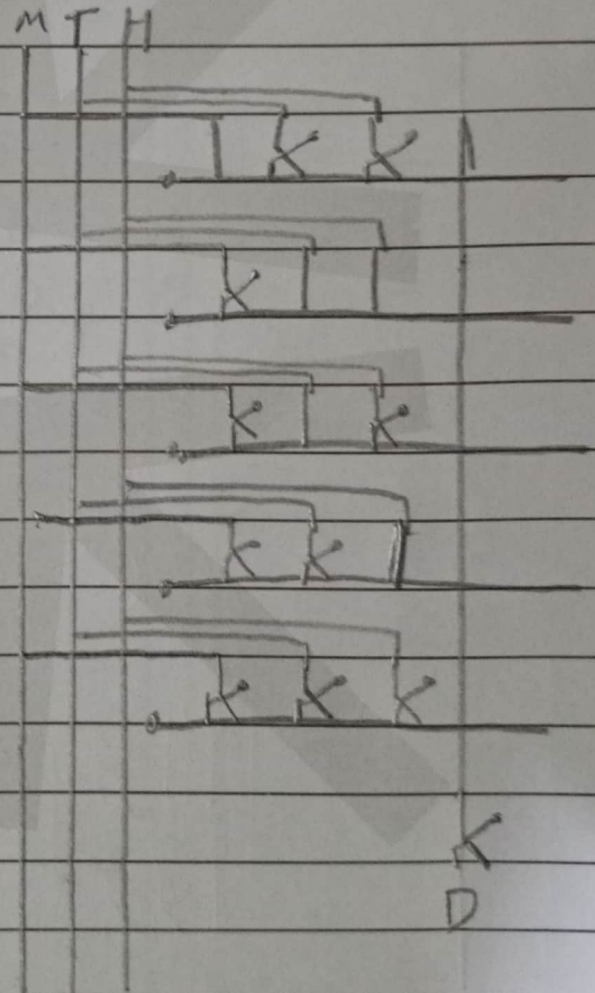
1 0 0 1 $M \cdot \bar{T} \cdot \bar{H}$

1 0 1 1 $M \cdot \bar{T} \cdot H$

1 1 0 1 $M \cdot T \cdot \bar{H}$

1 1 1 1 $M \cdot T \cdot H$

b) $D = M + (\bar{M} \cdot T \cdot H)$

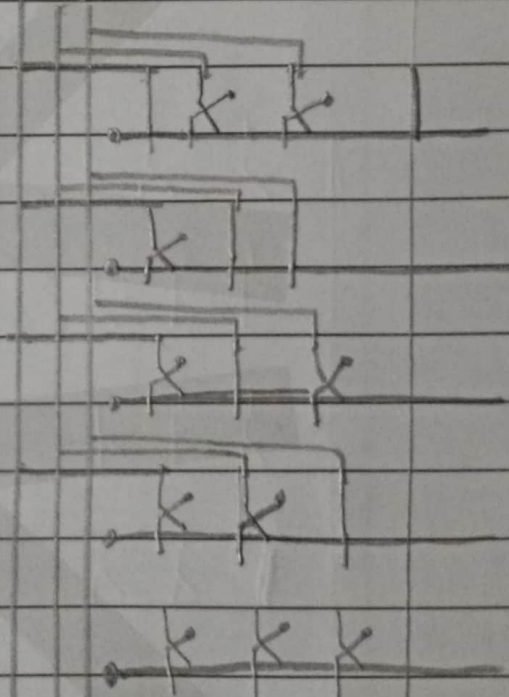


- Q6.a) $M \Rightarrow 1$ if foreign metal is detected
 $W \Rightarrow 1$ if package is greater than 5 kg
 $D \Rightarrow 1$ if package > 50 cm
 $R \Rightarrow 1$ if pneumatic arm pushes package

M	W	D	R
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

 $M \cdot W \cdot D$ $M \cdot \bar{W} \cdot \bar{D}$ $M \cdot \bar{W} \cdot D$ $M \cdot W \cdot \bar{D}$ $M \cdot W \cdot D$

MWD



X
R

$$R = M + (M \cdot W \cdot D)$$

Q7.a) $B \Rightarrow 1$ if it strikes an object

$L \Rightarrow 1$ if broken

$W \Rightarrow 1$ if weight > 1000 kg

$D \Rightarrow 1$ if the door reopens

B	L	W	D
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

$$\bar{B} \cdot L \cdot \bar{W}$$

$$\bar{B} \cdot L \cdot W$$

$$B \cdot \bar{L} \cdot \bar{W}$$

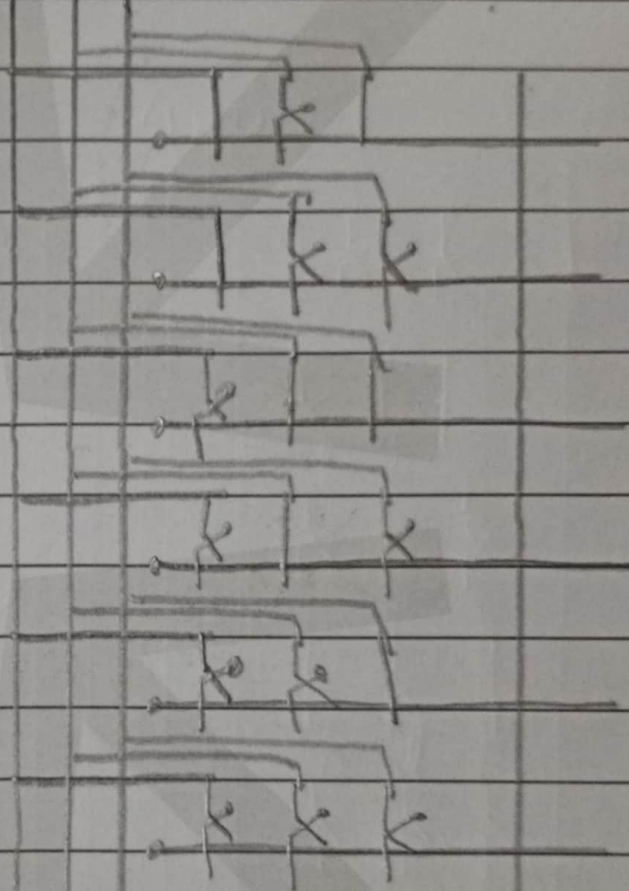
$$B \cdot \bar{L} \cdot W$$

$$\bar{B} \cdot L \cdot \bar{W}$$

$$B \cdot L \cdot W$$

b) $D = B + L + (W \cdot L)$

B L W



D

Q8.a) $M \Rightarrow 1$ if pulled

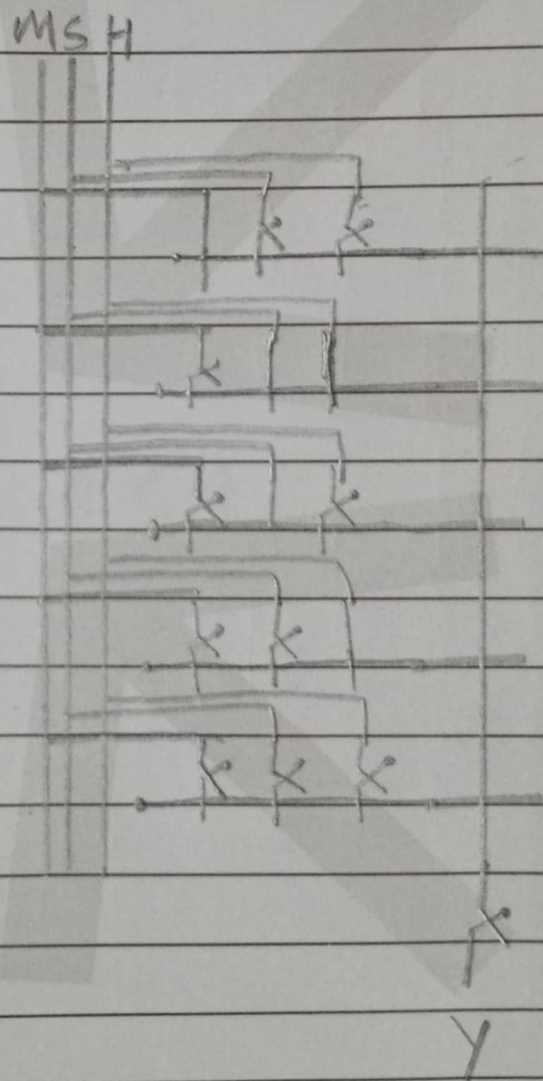
$S \Rightarrow 1$ if smoke opacity $> 20\%$

$H \Rightarrow 1$ if temperature $> 75^\circ\text{C}$

$Y \Rightarrow 1$ if activated

M	S	H	Y	
0	0	0	0	
0	0	1	0	
0	1	0	0	
0	1	1	1	$\bar{M} \cdot \bar{S} \cdot H$
1	0	0	1	$\bar{M} \cdot S \cdot \bar{H}$
1	0	1	1	$M \cdot \bar{S} \cdot H$
1	1	0	1	$M \cdot S \cdot \bar{H}$
1	1	1	1	$M \cdot S \cdot H$

b) $Y = M + (\bar{M} \cdot S \cdot H)$



Q9.a) $L \Rightarrow 1$ if daytime

$C \Rightarrow 1$ if all chickens are inside

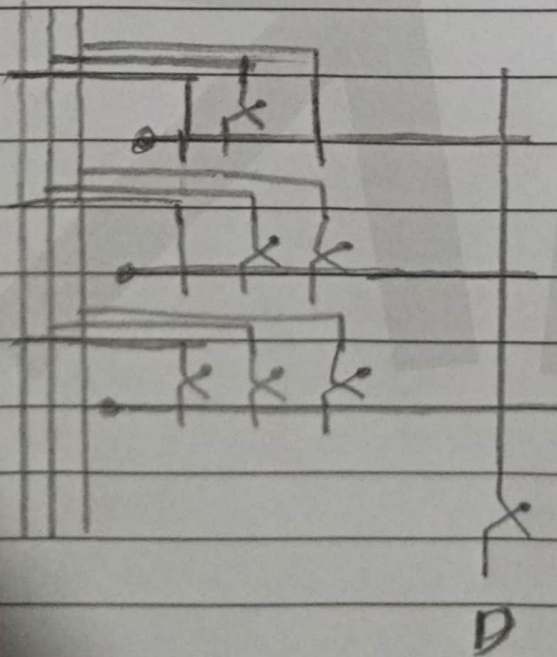
$R \Rightarrow 1$ if it is raining

$D \Rightarrow 1$ if the door should close

L	C	R	D	
0	0	0	0	
0	0	1	0	
0	1	0	1	$\bar{L} \cdot C \cdot \bar{R}$
0	1	1	1	$\bar{L} \cdot C \cdot R$
1	0	0	0	
1	0	1	0	
1	1	0	0	
1	1	1	1	$L \cdot C \cdot R$

b) $D = (\bar{L} \cdot C) + (L \cdot R \cdot C)$

L C R



Q10. a) $B \Rightarrow 1$ if battery $< 15\%$

$D \Rightarrow 1$ if distance $> 2 \text{ km}$

$W \Rightarrow 1$ if wind $> 20 \text{ km/h}$

$S \Rightarrow 1$ if activated

B	D	W	S
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

b) $S = (B \cdot D) + (W \cdot B)$

